Python Beginner Practice

A small starter pack to practise Python and upload your work to GitHub

How to use this PDF

1. Create a folder on your computer named **Python-Basics**.
2. Create one .py file for each program below (for example, **hello.py**).
3. Copy the matching code into the file and save it.
4. Run it with Python, check the output, and make sure you understand what each line does.
5. Upload the .py files to a public GitHub repository named **Python-Basics**.

Tip: You can practise in IDLE, VS Code, PyCharm, or an online editor such as Replit. Do not upload passwords, API keys, private data, or confidential college information.

1. Hello World — hello.py

Purpose: Prints simple messages.

```
print("Hello, GitHub!")
print("I am learning Python.")
```

2. Even or Odd — even_odd.py

Concepts: input, int conversion, if/else, remainder operator (%).

```
number = int(input("Enter a number: "))

if number % 2 == 0:
    print(number, "is even")
else:
    print(number, "is odd")
```

3. Simple Calculator — calculator.py

Concepts: numeric input, if/elif/else, arithmetic, basic error handling.

```
num1 = float(input("Enter first number: "))
operator = input("Enter an operator (+, -, *, /): ")
num2 = float(input("Enter second number: "))

if operator == "+":
    print("Result:", num1 + num2)
elif operator == "-":
    print("Result:", num1 - num2)
elif operator == "*":
    print("Result:", num1 * num2)
elif operator == "/":
    if num2 != 0:
        print("Result:", num1 / num2)
    else:
        print("Error: cannot divide by zero")
else:
    print("Invalid operator")
```

4. Factorial — factorial.py

Concepts: for loop, range, accumulator variable. Example: 5 factorial = 120.

```
number = int(input("Enter a non-negative integer: "))

if number < 0:
    print("Factorial is not defined for negative numbers")
```

```
else:
    result = 1
    for i in range(1, number + 1):
        result = result * i
    print("Factorial:", result)
```

5. Prime Number Check — prime.py

Concepts: loops, conditions, break. A prime number has exactly two positive factors.

```
number = int(input("Enter an integer: "))

if number <= 1:
    print(number, "is not prime")
else:
    is_prime = True
    for i in range(2, int(number ** 0.5) + 1):
        if number % i == 0:
            is_prime = False
            break

    if is_prime:
        print(number, "is prime")
    else:
        print(number, "is not prime")
```

6. Multiplication Table — multiplication.py

Concepts: for loop and formatted output.

```
number = int(input("Enter a number: "))

for i in range(1, 11):
    print(number, "x", i, "=", number * i)
```

Upload your files to GitHub

1. Sign in at github.com.
2. Open your repository named Python-Basics. If you have not created it, choose New repository, enter the name, select Public, and create it with a README.
3. Choose Add file → Upload files.
4. Select your .py files from the Python-Basics folder. You may upload all six together.
5. Enter a commit message such as: Add beginner Python practice programs.
6. Choose Commit changes.

Suggested README text

```
# Python Basics

This repository contains my beginner Python practice programs.

## Programs
- hello.py — prints a welcome message
- even_odd.py — checks whether a number is even or odd
- calculator.py — performs basic arithmetic
- factorial.py — calculates a factorial
- prime.py — checks whether a number is prime
- multiplication.py — prints a multiplication table

## About
I am learning Python and using this repository to practise programming fundamentals.
```

Before uploading: Run each program at least once, correct any errors, and make sure you can explain the logic. These are practice examples; personalise them as you learn.